



AGENTIC AI

Day 4

ReAct & ReWOO Architectures
AI Agent Frameworks



ARCHITECTURE • FRAMEWORKS • LOCAL LLM

Learning Objectives



Understand ReAct (Reasoning + Acting)



Understand ReWOO Architecture



Compare ReAct vs ReWOO



PRACTICALS

1

Install local LLM
using Ollama

2

Connect Ollama
with Python

AI Agents – Quick Recap



Think

Cognitive processing & planning



Act

Execute actions via tools



Decide

Iterative loop decision-making



Advanced

Goes beyond traditional AI

What is Agentic Architecture?



Structure of how
AI agent works



Defines the
decision-making process



Controls thinking
and execution



Enables intelligent
automation

ReAct Architecture Overview

ReAct

Reasoning + Acting

- Combines thinking and acting in a unified loop
- Dynamically decides which tool to call next
- Continuously adapts based on observations
- Used in state-of-the-art LLM agent systems



THINK → ACT → OBSERVE → REPEAT

ReAct: Thought → Action → Observation



↻ Repeat until Final Answer

ReAct Loop – In Depth



Thought

LLM decides the next step based on context & history



Action

Agent calls an external API, tool, or search engine



Observation

Result from the tool is fed back into the context



Loop Continues

Thought-Action-Observation repeats until resolved

ReAct in Action – Example

Scenario: 'What is the current weather in Chennai?'



Thought

I need live weather data. I should call a weather API.



Action

→ Call WeatherAPI('Chennai')



Observation

← Response: { temp: 35°C, condition: 'Sunny' }



Final Answer

The current weather in Chennai is 35°C and Sunny.

ReAct – Key Features



Dynamic Decision-Making

Adapts thinking based on real-time results



Real-Time Adaptation

Updates strategy after each observation



Tool Integration

Seamlessly calls APIs, search, databases



High Flexibility

Handles diverse, open-ended tasks

ReAct – Limitations

01



Slower Execution

Multiple loop iterations take more time

02



Higher Computational Cost

Each LLM call adds to token usage & cost

03



Multiple Iterations Required

Complex tasks may loop many times before resolving



QUIZ

ReAct Architecture

What does ReAct stand for?

A) Real Actions

B) Reasoning + Acting

C) Recursive Automation

D) Reinforced Actions

ANSWER

What does ReAct stand for?

B) Reasoning + Acting

ReAct combines Reasoning (thinking step-by-step) with Acting (using tools/APIs) in a continuous loop to solve complex tasks.

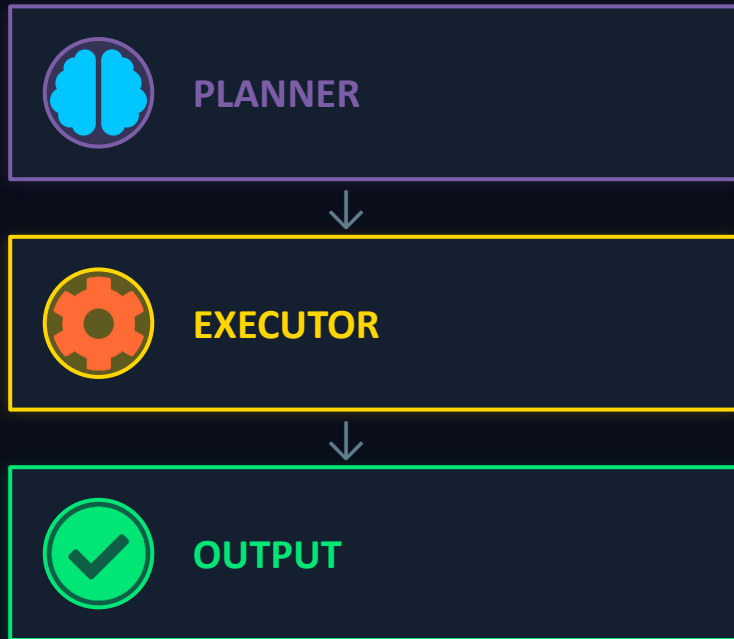


ReWOO Architecture Overview

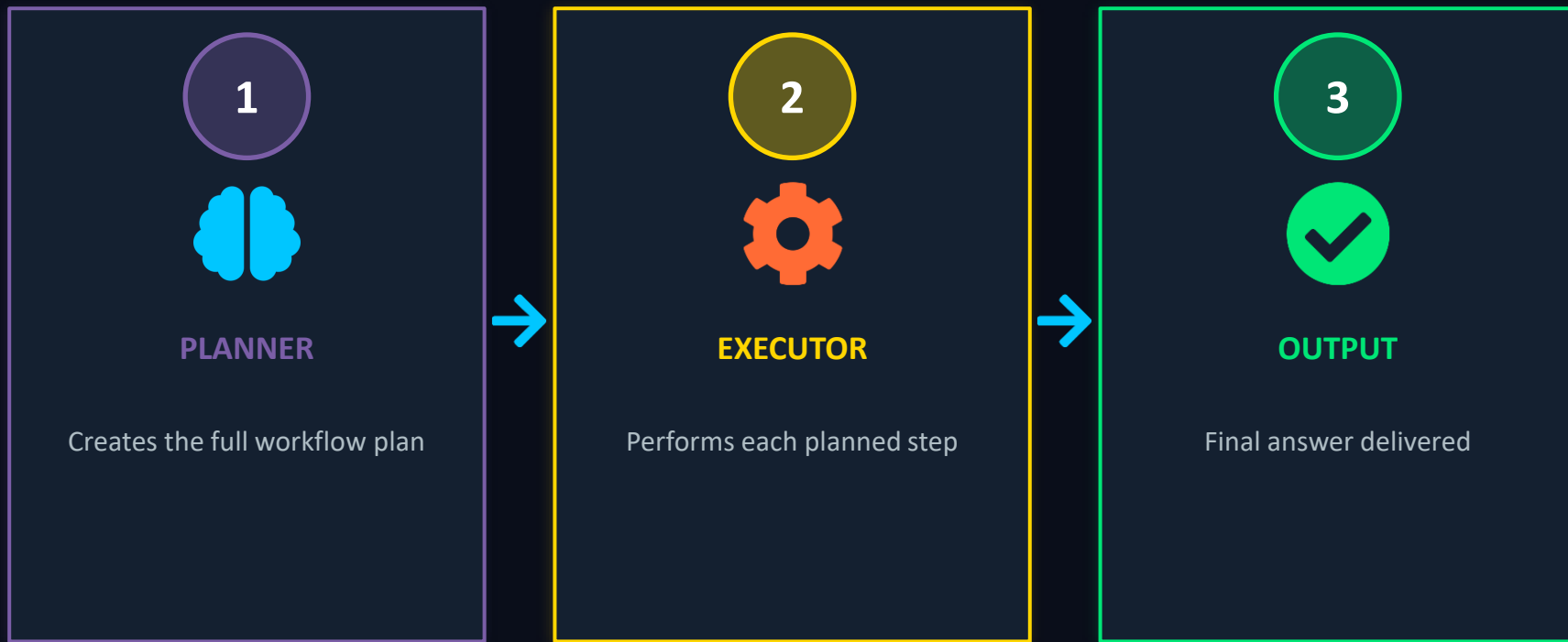
ReWOO

*Reasoning Without
Observation*

- Plans the entire workflow upfront before execution
- Executes all steps sequentially without looping
- Efficient and cost-effective for structured tasks
- Predictable and deterministic output



ReWOO: Plan → Execute → Output



No mid-step observation needed — plan is executed fully

ReWOO – 4 Step Process

1

Understand Task

Parse user query and define objective clearly

2

Create Full Workflow

Planner creates a complete step-by-step roadmap

3

Execute Sequentially

Executor runs all steps one by one without feedback loops

4

Deliver Result

Final answer synthesized and returned to user

ReWOO in Action – Example

Scenario: 'Research and summarize the top 3 AI trends in 2025'



Plan

Step 1: Search web for AI trends 2025



Plan

Step 2: Extract top 3 trends from results



Execute

Step 3: Run web search tool → Fetch articles



Output

Step 4: Summarize 3 key trends with descriptions

ReWOO – Key Features



Faster Execution

Single-pass execution — no repeated loops



Cost-Efficient

Fewer LLM calls = lower token usage



Structured Workflow

Well-defined, predictable step sequence



Predictable Behavior

Consistent output for similar task inputs

ReWOO – Limitations

01



No Mid-Step Correction

Cannot adjust if an intermediate step fails or yields bad data

02



Less Flexible

Struggles with dynamic, evolving tasks that require re-planning

03



Cannot Adapt Dynamically

All decisions made upfront — no course corrections mid-execution



QUIZ

ReWOO Architecture

Which architecture plans BEFORE executing all steps?

A) ReAct

B) ReWOO

C) Both equally

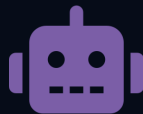
D) Neither

ANSWER

Which architecture plans BEFORE executing?

B) ReWOO

ReWOO (Reasoning Without Observation) creates a complete plan upfront, then executes all steps sequentially without mid-loop feedback.



ReAct vs ReWOO – Comparison

Feature	ReAct	ReWOO
Approach	Loop-Based	Plan-Based
Flexibility	High	Medium
Speed	Slower	Faster
Cost	Higher	Lower
Adaptability	Dynamic	Structured

ReAct vs ReWOO – Key Differences



ReAct

- Thinks while solving the task
- Observes tool results and adapts
- Dynamic — good for open-ended problems
- Higher token usage per task



ReWOO

- Plans everything before solving
- Executes steps without intermediate feedback
- Structured — ideal for well-defined workflows
- Lower cost, higher execution speed

What is a Local LLM?



Runs on Local System

Model runs directly on your machine — no cloud required



Data Stays Private

No data leaves your machine — complete privacy



No Internet Required

Works fully offline once model is downloaded



No API Dependency

Use powerful LLMs without paid external APIs

Ollama – Run LLMs Locally

Run LLM
locally

Supports multiple
models (Llama, Mistral)



Easy to install
& use

Python / REST
API ready

```
$ ollama run llama3
```

Why Use a Local LLM?



Privacy & Security

Sensitive data stays on your machine — never sent to cloud



Cost Saving

Zero API charges — run as many queries as you want



Offline Capability

Fully functional without internet connectivity



Faster Local Processing

Low latency responses — no network round-trip

QUIZ

Local LLM • Question 1 of 1

Which tool is used to run LLMs locally?

A) Hugging Face

B) LangChain

C) Ollama

D) OpenAI



ANSWER

Which tool is used to run LLMs locally?

C) Ollama

Ollama lets you download and run powerful open-source LLMs (like Llama 3, Mistral) directly on your laptop — no API key needed!



Session Summary



ReAct

Dynamic, flexible, loop-based reasoning



ReWOO

Fast, structured, plan-based execution



Local LLM

Secure, offline, cost-effective AI



Practical Setup

Ollama + Python + IBM Cloud API Key

Now Let's Get Hands-On!

01



Install Ollama on your local machine

02



Connect Ollama with Python scripts

03



Generate an IBM Cloud API Key

04



Build your first ReAct/ReWOO agent